SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I G. Santosh Kumar *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Santosh Kumar

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%				
Identification of Key Issues	20%				
Application of Concepts	25%				
Decision Making	20%				
Clarity of Response	15%				

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses.**
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation.**
2. Expand the compressed address back into its **full hexadecimal representation.**
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range.**
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understandin g of the	Shows good understandi ng with minor gaps	Partial understandin g; misses key aspects	Misinterpret s or fails to understand the scenario

		scenario and context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

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CO2-AT2 - Scenario Based

1. Smart Campus network (192.168.100.0/24)

Given network : 192.168.100.0/24

School of computing C(25)

- Subnet Mask : 255.255.255.128
- Network address : 192.168.100.0
- First Host : 192.168.100.1
- Last Host : 192.168.100.126
- Broad cast address : 192.168.100.127
- Usable Hosts : $2^7 - 2 = 126$

School of Electronics C(26)

Next available Subnet:

- Network address : 192.168.100.128
- Subnet Mask : 255.255.255.192
- First Host : 192.168.100.129
- Last Host : 192.168.100.190
- Broadcast address : 192.168.100.191
- Usable Hosts : $2^6 - 2 = 62$

School of Mechanical Engineering (IIT)

Next available subnet:

- Network address: 192.168.100.192
- Subnet Mask: 255.255.255.224
- First Host: 192.168.100.193
- Last Host: 192.168.100.222
- Broadcast address: 192.168.100.223
- Usable Hosts: $2^5 - 2 = 30$

Remaining unused IP range

Allocated up to:

192.168.100.223

Remaining unused:

192.168.100.224 – 192.168.100.255

2. Software Company (10.10.0.0/16)

Software Development (/24)

- Network address: 10.10.0.0
- Host Range: 10.10.0.1 – 10.10.0.254
- Broadcast: 10.10.0.255
- Usable Hosts: 254

Quality assurance (/25)

Next Subnet:

- Network address: 10.10.1.0
- Host range: 10.10.1.1 – 10.10.1.126
- Broadcast: 10.10.1.127
- Usable Hosts: 126

Human Resources (/26)

Next subnet:

- Network address: 10.10.1.128
- Host Range: 10.10.1.129 – 10.10.1.190
- Broadcast: 10.10.1.191
- Usable Hosts: 62

Executive Management (/27)

Next subnet:

- Network address: 10.10.1.192
- Host Range: 10.10.1.193 – 10.10.1.222
- Broadcast: 10.10.1.223
- Usable Hosts: 30

Remaining unused Range

Allocated

10.10.0.0 – 10.10.1.223

Remaining

10.10.1.224 – 10.10.255.255

3. IPv6 Hospital Network

Given:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Compressed form

2001:db8:abcd::/64

2. Expanded form

2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network prefix (/64)

2001:db8:abcd:0::/64

4 & 5. Number of Host addresses

Host bits = 128 − 64 = 64.

2^{64} = 18,446,744,073,709,551,616 host addresses

Routing table

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination IP:

192.168.2.45

5. Company LAN's

Administration LAN

Network:

192.168.10.0 /26

- Network address : 192.168.10.0
- first Host : 192.168.10.1
- Last Host : 192.168.10.62
- Broadcast : 192.168.10.63
- Gateway : 192.168.10.1

Three Example PC

- PC1 → 192.168.10
- PC2 → 192.168.10
- PC3 → 192.168.10

Finance LAN

Network:

192.168.10.64/26

- Network address : 192.168.10.64
- first Host : 192.168.10.65
- Last Host : 192.168.10.126
- Broadcast : 192.168.10.127

· Gateway : 192.168.10.65

Three Example PCs

· PC1 → 192.168.10.66

· PC2 → 192.168.10.67

· PC3 → 192.168.10.68